

Biswajit Mandal

biswajit328328@gmail.com | linkedin.com/in/biswajit-mandal328 | github.com/biswajit328

Third-year Computer Engineering student at Thapar University, seeking a Software Development role. Proficient in Python, C++, and JavaScript with practical experience in backend development, REST APIs, object-oriented design, distributed systems, and databases (SQL and NoSQL) — applied across three independently built, production-deployed applications.

EDUCATION

Thapar Institute of Engineering and Technology B.Tech in Computer Engineering	Patiala, Punjab Aug. 2024 – Jun. 2028
---	--

PROJECTS

Nexora – AI-Powered Campus Opportunity Recommendation Platform <i>Github</i>	Apr. – Present
- FastAPI, PostgreSQL, SQLAlchemy, Alembic, React, TypeScript, Tailwind CSS, Google Gemini API, Docker, GitHub Actions - Built a full-stack platform ranking internships, hackathons, and scholarships by skill match, eligibility, interests, deadlines, and preferences, with JWT authentication and student/organizer/admin roles. - Designed a normalized 14-table PostgreSQL schema with race-safe, case-insensitive uniqueness using citext and atomic ON CONFLICT inserts. - Developed a deterministic, explainable recommendation engine with a three-state eligibility filter and weighted scoring; achieved +0.52 NDCG@3 over a deadline-only baseline. - Built a Gemini-powered ingestion pipeline converting unstructured opportunity announcements into schema-constrained data, with independent semantic validation and a provider-agnostic, mock-testable AI abstraction. - Implemented the complete organizer submission workflow from self-service onboarding and raw-text submission to AI-assisted admin review and approval, backed by 74+ pytest tests and GitHub Actions CI.	
Neural Predictive Maintenance System – Smart Factory AI <i>Github</i>	Apr. – May 2026
- Python, TensorFlow/Keras, FastAPI, scikit-learn, Docker Compose, PostgreSQL, Redis, MQTT, GitHub Actions - Designed and shipped a real-time ML system ingesting 9 industrial sensor streams to predict machine failure probability (ROC-AUC/PR-AUC evaluated), using a neural temporal sensor-fusion model with 3 parallel Conv1D branches in Keras alongside a leakage-free Random Forest baseline. - Developed 6 RESTful FastAPI endpoints for live inference, stream simulation, health checks, and diagnostics; containerized the system with Docker Compose across 8 services, enforced by full automated test coverage via GitHub Actions CI/CD.	
Hospital Drug Shortage Simulation – Open-Source RL Benchmark <i>Github</i>	Mar. – Apr. 2026
- Python, Reinforcement Learning, Hugging Face Spaces, FastAPI, Distributed Simulation - Architected a distributed multi-agent RL environment spanning 6 task families and 3 difficulty tiers, evaluating AI decision-making under budget, supply-chain, and clinical constraints, with standardized action interfaces and deterministic graders. - Deployed live on Hugging Face Spaces with a validator-safe RESTful API contract, publicly used by the AI research community for reproducible benchmarking.	

RELEVANT COURSEWORK

- Data Structures & Algorithms - Object-Oriented Programming - Operating Systems	- Database Management Systems (DBMS) - Machine Learning - Software Engineering
--	--

SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL
Frameworks/Tools: FastAPI, React, Tailwind CSS, TensorFlow/Keras, scikit-learn, Docker, GitHub Actions, AWS, Linux
Databases: PostgreSQL, MySQL, MongoDB
AI/ML: Google Gemini API, TensorFlow/Keras, scikit-learn, pandas, NumPy
Practices: REST API design, Object-oriented design, automated testing, CI/CD, code reviews

ACHIEVEMENTS

CodeChef Certified – Design & Analysis of Algorithms (May 2026), covering dynamic programming, graph algorithms, complexity analysis, and optimization. - Independently built and publicly deployed 3 software and AI/ML systems beyond coursework, spanning recommendation systems, predictive maintenance, and reinforcement-learning benchmarking. - Built projects with production-grade engineering practices, including automated testing, Docker-based deployments, REST APIs, CI/CD pipelines, and reproducible evaluation workflows.
--